Neural Computation -- 1990

51 article(s), all issues

1. Generalized Deformable Models, Statistical Physics, and Matching Problems

Alan Yuille

https://doi.org/10.1162/neco.1990.2.1.1 | pp. 1-24

We describe how to formulate matching and combinatorial problems of vision and neural network theory by generalizing elastic and deformable templates models to include binary matching elements. Techniques from statistical physics, which can be interpreted as computing marginal probability distributions, are then used to analyze these models and are shown to (1) relate them to existing theories and (2) give insight into the relations between, and relative effectivenesses of, existing theories. In particular we exploit the power of statistical techniques to put global constraints on the set of allowable states of the binary matching elements. The binary elements can then be removed analytically before minimization. This is demonstrated to be preferable to existing methods of imposing such constraints by adding bias terms in the energy functions. We give applications to winner-take-all networks, correspondence for stereo and long-range motion, the traveling salesman problem, deformable template matching, learning, content addressable memories, and models of brain development. The biological plausibility of these networks is briefly discussed.

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2. An Optoelectronic Architecture for Multilayer Learning in a Single Photorefractive Crystal

Carsten Peterson, Stephen R. Redfield, James D. Keeler, Eric Hartman

https://doi.org/10.1162/neco.1990.2.1.25 | pp. 25-34

We propose a simple architecture for implementing supervised neural network models optically with photorefractive technology. The architecture is very versatile: a wide range of supervised learning algorithms can be implemented including mean-field-theory, backpropagation, and Kanerva-style networks. Our architecture is based on a single crystal with spatial multiplexing rather than the more commonly used angular multiplexing. It handles hidden units and places no restrictions on connectivity. Associated with spatial multiplexing are certain physical phenomena, rescattering and beam depletion, which tend to degrade the matrix multiplications. Detailed simulations including beam absorption and grating decay show that the supervised learning algorithms (slightly modified) compensate for these degradations.

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3. VLSI Implementation of Neural Classifiers

Arun Rao, M. Walker, Lawrence T. Clark, L.A. Akers, R. Grondin

https://doi.org/10.1162/neco.1990.2.1.35 | pp. 35-43

The embedding of neural networks in real-time systems performing classification and clustering tasks requires that models be implemented in hardware. A flexible, pipelined associative memory capable of operating in real-time is proposed as a hardware substrate for the emulation of neural fixed-radius clustering and binary classification schemes. This paper points out several important considerations in the

development of hardware implementations. As a specific example, it is shown how the ART1 paradigm can be functionally emulated by the limited resolution pipelined architecture, in the absence of full parallelism.

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4. Coherent Compound Motion: Corners and Nonrigid Configurations

Steven W. Zucker, Lee Iverson, Robert A. Hummel

https://doi.org/10.1162/neco.1990.2.1.44 | pp. 44-57

Consider two wire gratings, superimposed and moving across each other. Under certain conditions the two gratings will cohere into a single, compound pattern, which will appear to be moving in another direction. Such coherent motion patterns have been studied for sinusoidal component gratings, and give rise to percepts of rigid, planar motions. In this paper we show how to construct coherent motion displays that give rise to nonuniform, nonrigid, and nonplanar percepts. Most significantly, they also can define percepts with corners. Since these patterns are more consistent with the structure of natural scenes than rigid sinusoidal gratings, they stand as interesting stimuli for both computational and physiological studies. To illustrate, our display with sharp corners (tangent discontinuities or singularities) separating regions of coherent motion suggests that smoothing does not cross tangent discontinuities, a point that argues against existing (regularization) algorithms for computing motion. This leads us to consider how singularities can be confronted directly within optical flow computations, and we conclude with two hypotheses: (1) that singularities are represented within the motion system as multiple directions at the same retinotopic location; and (2) for component gratings to cohere, they must be at the same depth from the viewer. Both hypotheses have implications for the neural computation of coherent motion.

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5. A Complementarity Mechanism for Enhanced Pattern Processing

James L. Adams

https://doi.org/10.1162/neco.1990.2.1.58 | pp. 58-70

The parallel ON- and OFF-center signals flowing from retina to brain suggest the operation of a complementarity mechanism. This paper shows what such a mechanism can do in higher-level visual processing. In the proposed mechanism, inhibition and excitation, both feedforward, coequally compete within each hierarchical level to discriminate patterns. A computer model tests complementarity in the context of an adaptive, self-regulating system. Three other mechanisms (gain control, cooperativity, and adaptive error control) are included in the model but are described only briefly. Results from simulations show that complementarity markedly improves both speed and accuracy in pattern learning and recognition. This mechanism may serve not only vision but other types of processing in the brain as well.

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6. Hebb-Type Dynamics is Sufficient to Account for the Inverse Magnification Rule in Cortical Somatotopy

Kamil A. Grajski, Michael M. Merzenich

https://doi.org/10.1162/neco.1990.2.1.71 | pp. 71-84

The inverse magnification rule in cortical somatotopy is the experimentally derived inverse relationship between cortical magnification (area of somatotopic map representing a unit area of skin surface) and

receptive field size (area of restricted skin surface driving a cortical neuron). We show by computer simulation of a simple, multilayer model that Hebb-type synaptic modification subject to competitive constraints is sufficient to account for the inverse magnification rule.

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7. Optimal Plasticity from Matrix Memories: What Goes Up Must Come Down

David Willshaw, Peter Dayan

https://doi.org/10.1162/neco.1990.2.1.85 | pp. 85-93

A recent article (Stanton and Sejnowski 1989) on long-term synaptic depression in the hippocampus has reopened the issue of the computational efficiency of particular synaptic learning rules (Hebb 1949; Palm 1988a; Morris and Willshaw 1989) — homosynaptic versus heterosynaptic and monotonic versus nonmonotonic changes in synaptic efficacy. We have addressed these questions by calculating and maximizing the signal-to-noise ratio, a measure of the potential fidelity of recall, in a class of associative matrix memories. Up to a multiplicative constant, there are three optimal rules, each providing for synaptic depression such that positive and negative changes in synaptic efficacy balance out. For one rule, which is found to be the Stent-Singer rule (Stent 1973; Rauschecker and Singer 1979), the depression is purely heterosynaptic; for another (Stanton and Sejnowski 1989), the depression is purely homosynaptic; for the third, which is a generalization of the first two, and has a higher signal-to-noise ratio, it is both heterosynaptic and homosynaptic. The third rule takes the form of a covariance rule (Sejnowski 1977a,b) and includes, as a special case, the prescription due to Hopfield (1982) and others (Willshaw 1971; Kohonen 1972).

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8. Pattern Segmentation in Associative Memory

DeLiang Wang, Joachim M. Buhmann, Christoph von der Malsburg

https://doi.org/10.1162/neco.1990.2.1.94 | pp. 94-106

The goal of this paper is to show how to modify associative memory such that it can discriminate several stored patterns in a composite input and represent them simultaneously. Segmention of patterns takes place in the temporal domain, components of one pattern becoming temporally correlated with each other and anticorrelated with the components of all other patterns. Correlations are created naturally by the usual associative connections. In our simulations, temporal patterns take the form of oscillatory bursts of activity. Model oscillators consist of pairs of local cell populations connected appropriately. Transition of activity from one pattern to another is induced by delayed self-inhibition or simply by noise.

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9. A Neural Net Associative Memory for Real-Time Applications

Gregory L. Heileman, Giorgos Papadourakis, Michael Georgiopoulos

https://doi.org/10.1162/neco.1990.2.1.107 | pp. 107-115

A parallel hardware implementation of the associative memory neural network introduced by Hopfield is described. The design utilizes the Geometric Arithmetic Parallel Processor (GAPP), a commercially available single-chip VLSI general-purpose array processor consisting of 72 processing elements. The ability to cascade these chips allows large arrays of processors to be easily constructed and used to implement the Hopfield network. The memory requirements and processing times of such arrays are

analyzed based on the number of nodes in the network and the number of exemplar patterns. Compared with other digital implementations, this design yields significant improvements in runtime performance and offers the capability of using large neural network associative memories in real-time applications.

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10. Gram—Schmidt Neural Nets

S.J. Orfanidis

https://doi.org/10.1162/neco.1990.2.1.116 | pp. 116-126

A new type of feedforward multilayer neural net is proposed that exhibits fast convergence properties. It is defined by inserting a fast adaptive Gram-Schmidt preprocessor at each layer, followed by a conventional linear combiner-sigmoid part which is adapted by a fast version of the backpropagation rule. The resulting network structure is the multilayer generalization of the gradient adaptive lattice filter and the Gram-Schmidt adaptive array.

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11. Errata

(authors unknown)

https://doi.org/10.1162/neco.1990.2.1.127 | pp. 127-127

No abstract available.

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12. Visual Perception of Three-Dimensional Motion

David J. Heeger, Allan Jepson

https://doi.org/10.1162/neco.1990.2.2.129 | pp. 129-137

As an observer moves and explores the environment, the visual stimulation in his eye is constantly changing. Somehow he is able to perceive the spatial layout of the scene, and to discern his movement through space. Computational vision researchers have been trying to solve this problem for a number of years with only limited success. It is a difficult problem to solve because the relationship between the optical-flow field, the 3D motion parameters, and depth is nonlinear. We have come to understand that this nonlinear equation describing the optical-flow field can be split by an exact algebraic manipulation to yield an equation that relates the image velocities to the translational component of the 3D motion alone. Thus, the depth and the rotational velocity need not be known or estimated prior to solving for the translational velocity. The algorithm applies to the general case of arbitrary motion with respect to an arbitrary scene. It is simple to compute and it is plausible biologically.

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13. Distributed Symbolic Representation of Visual Shape

Eric Saund

https://doi.org/10.1162/neco.1990.2.2.138 | pp. 138-151

The notion of distributed representation has gained significance in explanations of connectionist or “neural” networks. This communication shows that the concept also offers motivation in devising representations for

visual shape within a symbolic computing paradigm. In a representation for binary (silhouette) shapes, and in analogy with conventional distributed connectionist networks, descriptive power is gained when microfeatures are available naming important spatial relationships in images. Our symbolic approach is introduced through a vocabulary of 31 “hand built” shape descriptors operating in the two-dimensional shape world of the dorsal fins of fishes.

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14. Modeling Orientation Discrimination at Multiple Reference Orientations with a Neural Network

M. Devos, Guy A. Orban

https://doi.org/10.1162/neco.1990.2.2.152 | pp. 152-161

We trained a multilayer perceptron with backpropagation to perform stimulus orientation discrimination at multiple references using biologically plausible values as input and output. Hidden units are necessary for good performance only when the network must operate at multiple reference orientations. The orientation tuning curves of the hidden units change with reference. Our results suggest that at least for simple parameter discriminations such as orientation discrimination, one of the main functions of further processing in the visual system beyond striate cortex is to combine signals representing stimulus and reference.

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15. Temporal Differentiation and Violation of Time-Reversal Invariance in Neurocomputation of Visual Information

D. S. Tang, Vinod Menon

https://doi.org/10.1162/neco.1990.2.2.162 | pp. 162-172

Information-theoretic techniques have been employed to study the time-dependent connection strength of a three-layer feedforward neural network. The analysis shows (1) there is a natural emergence of time-dependent receptive field that performs temporal differentiation and (2) the result is shown to be a consequence of a mechanism based on violation of the time-reversal invariance in the visual information processing system. Both analytic and numerical studies are presented.

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16. Analysis of Linsker's Simulations of Hebbian Rules

David Mackay, Kenneth D. Miller

https://doi.org/10.1162/neco.1990.2.2.173 | pp. 173-187

Linsker has reported the development of center-surround receptive fields and oriented receptive fields in simulations of a Hebb-type equation in a linear network. The dynamics of the learning rule are analyzed in terms of the eigenvectors of the covariance matrix of cell activities. Analytic and computational results for Linsker's covariance matrices, and some general theorems, lead to an explanation of the emergence of center-surround and certain oriented structures. We estimate criteria for the parameter regime in which center-surround structures emerge.

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17. Generalizing Smoothness Constraints from Discrete Samples

Chuanyi Ji, Robert R. Snapp, Demetri Psaltis

https://doi.org/10.1162/neco.1990.2.2.188 | pp. 188-197

We study how certain smoothness constraints, for example, piecewise continuity, can be generalized from a discrete set of analog-valued data, by modifying the error backpropagation, learning algorithm. Numerical simulations demonstrate that by imposing two heuristic objectives — (1) reducing the number of hidden units, and (2) minimizing the magnitudes of the weights in the network — during the learning process, one obtains a network with a response function that smoothly interpolates between the training data.

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18. The Upstart Algorithm: A Method for Constructing and Training Feedforward Neural Networks

Marcus Frean

https://doi.org/10.1162/neco.1990.2.2.198 | pp. 198-209

A general method for building and training multilayer perceptrons composed of linear threshold units is proposed. A simple recursive rule is used to build the structure of the network by adding units as they are needed, while a modified perceptron algorithm is used to learn the connection strengths. Convergence to zero errors is guaranteed for any boolean classification on patterns of binary variables. Simulations suggest that this method is efficient in terms of the numbers of units constructed, and the networks it builds can generalize over patterns not in the training set.

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19. Layered Neural Networks with Gaussian Hidden Units as Universal Approximations

Eric Hartman, James D. Keeler, J. Kowalski

https://doi.org/10.1162/neco.1990.2.2.210 | pp. 210-215

A neural network with a single layer of hidden units of gaussian type is proved to be a universal approximator for real-valued maps defined on convex, compact sets of R n .

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20. A Neural Network for Nonlinear Bayesian Estimation in Drug Therapy

Reza Shadmehr, David Z. D'Argenio

https://doi.org/10.1162/neco.1990.2.2.216 | pp. 216-225

The feasibility of developing a neural network to perform nonlinear Bayesian estimation from sparse data is explored using an example from clinical pharmacology. The problem involves estimating parameters of a dynamic model describing the pharmacokinetics of the bronchodilator theophylline from limited plasma concentration measurements of the drug obtained in a patient. The estimation performance of a backpropagation trained network is compared to that of the maximum likelihood estimator as well as the maximum a posteriori probability estimator. In the example considered, the estimator prediction errors (model parameters and outputs) obtained from the trained neural network were similar to those obtained using the nonlinear Bayesian estimator.

21. Analysis of Neural Networks with Redundancy

Yoshio Izui, Alex Pentland

https://doi.org/10.1162/neco.1990.2.2.226 | pp. 226-238

Biological systems have a large degree of redundancy, a fact that is usually thought to have little effect beyond providing reliable function despite the death of individual neurons. We have discovered, however, that redundancy can qualitatively change the computations carried out by a network. We prove that for both feedforward and feedback networks the simple duplication of nodes and connections results in more accurate, faster, and more stable computation.

22. Stability of the Random Neural Network Model

Erol Gelenbe

https://doi.org/10.1162/neco.1990.2.2.239 | pp. 239-247

In a recent paper (Gelenbe 1989) we introduced a new neural network model, called the Random Network, in which “negative” or “positive” signals circulate, modeling inhibitory and excitatory signals. These signals can arrive either from other neurons or from the outside world: they are summed at the input of each neuron and constitute its signal potential. The state of each neuron in this model is its signal potential, while the network state is the vector of signal potentials at each neuron. If its potential is positive, a neuron fires, and sends out signals to the other neurons of the network or to the outside world. As it does so its signal potential is depleted. We have shown (Gelenbe 1989) that in the Markovian case, this model has product form, that is, the steady-state probability distribution of its potential vector is the product of the marginal probabilities of the potential at each neuron. The signal flow equations of the network, which describe the rate at which positive or negative signals arrive at each neuron, are nonlinear, so that their existence and uniqueness are not easily established except for the case of feedforward (or backpropagation) networks (Gelenbe 1989). In this paper we show that whenever the solution to these signal flow equations exists, it is unique. We then examine two subclasses of networks — balanced and damped networks — and obtain stability conditions in each case. In practical terms, these stability conditions guarantee that the unique solution can be found to the signal flow equations and therefore that the network has a well-defined steady-state behavior.

23. The Perceptron Algorithm is Fast for Nonmalicious Distributions

Eric B. Baum

https://doi.org/10.1162/neco.1990.2.2.248 | pp. 248-260

Within the context of Valiant's protocol for learning, the perceptron algorithm is shown to learn an arbitrary half-space in time O(n 2 /■ 3) if D, the probability distribution of examples, is taken uniform over the unit sphere S n . Here ■ is the accuracy parameter. This is surprisingly fast, as “standard” approaches involve solution of a linear programming problem involving Ω(n/■) constraints in n dimensions. A modification of Valiant's distribution-independent protocol for learning is proposed in which the distribution and the function to be learned may be chosen by adversaries, however these adversaries may not communicate. It is argued that this definition is more reasonable and applicable to real world learning than Valiant's. Under this

definition, the perceptron algorithm is shown to be a distribution-independent learning algorithm. In an appendix we show that, for uniform distributions, some classes of infinite V-C dimension including convex sets and a class of nested differences of convex sets are learnable.

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24. Parallel Distributed Approaches to Combinatorial Optimization: Benchmark Studies on Traveling Salesman Problem

Carsten Peterson

https://doi.org/10.1162/neco.1990.2.3.261 | pp. 261-269

We present and summarize the results from 50-, 100-, and 200-city TSP benchmarks presented at the 1989 Neural Information Processing Systems (NIPS) postconference workshop using neural network, elastic net, genetic algorithm, and simulated annealing approaches. These results are also compared with a state-of-the-art hybrid approach consisting of greedy solutions, exhaustive search, and simulated annealing.

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25. Faster Learning for Dynamic Recurrent Backpropagation

Yan Fang, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1990.2.3.270 | pp. 270-273

September 01 1990 Faster Learning for Dynamic Recurrent Backpropagation Yan Fang, Yan Fang The Salk Institute, Computational Neurobiology Laboratory, 10010 N. Torrey Pines Road, La Jolla, CA 92037 USA Search for other works by this author on: This Site Google Scholar Terrence J. Sejnowski Terrence J. Sejnowski The Salk Institute, Computational Neurobiology Laboratory, 10010 N. Torrey Pines Road, La Jolla, CA 92037 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Yan Fang The Salk Institute, Computational Neurobiology Laboratory, 10010 N. Torrey Pines Road, La Jolla, CA 92037 USA Terrence J. Sejnowski The Salk Institute, Computational Neurobiology Laboratory, 10010 N. Torrey Pines Road, La Jolla, CA 92037 USA Received: January 08 1990 Accepted: May 23 1990 Online Issn: 1530-888X Print Issn: 0899-7667 © 1990 Massachusetts Institute of Technology1990 Neural Computation (1990) 2 (3): 270–273. https://doi.org/10.1162/neco.1990.2.3.270 Article history Received: January 08 1990 Accepted: May 23 1990 Cite Icon Cite Permissions Share Icon Share Twitter LinkedIn Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Yan Fang, Terrence J. Sejnowski; Faster Learning for Dynamic Recurrent Backpropagation. Neural Comput 1990; 2 (3): 270–273. doi: https://doi.org/10.1162/neco.1990.2.3.270 Download citation file: Ris (Zotero) Reference Manager EasyBib Bookends Mendeley Papers EndNote RefWorks BibTex toolbar search Search Dropdown Menu nav search search input Search input auto suggest search filter All ContentAll JournalsNeural Computation Search Advanced Search This content is only available as a PDF. © 1990 Massachusetts Institute of Technology1990 Article PDF first page preview Close Modal You do not currently have access to this content.

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26. A Dynamic Neural Network Model of Sensorimotor Transformations in the Leech

Shawn R. Lockery, Yan Fang, Terrence J. Sejnowski

https://doi.org/10.1162/neco.1990.2.3.274 | pp. 274-282

Interneurons in leech ganglia receive multiple sensory inputs and make synaptic contacts with many motor neurons. These “hidden” units coordinate several different behaviors. We used physiological and anatomical constraints to construct a model of the local bending reflex. Dynamic networks were trained on experimentally derived input-output patterns using recurrent backpropagation. Units in the model were modified to include electrical synapses and multiple synaptic time constants. The properties of the hidden units that emerged in the simulations matched those in the leech. The model and data support distributed rather than localist representations in the local bending reflex. These results also explain counterintuitive aspects of the local bending circuitry.

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27. Control of Neuronal Output by Inhibition at the Axon Initial Segment

Rodney J. Douglas, Kevan A Martin

https://doi.org/10.1162/neco.1990.2.3.283 | pp. 283-292

We examine the effect of inhibition on the axon initial segment (AIS) by the chandelier (“axoaxonic”) cells, using a simplified compartmental model of actual pyramidal neurons from cat visual cortex. We show that within generally accepted ranges, inhibition at the AIS cannot completely prevent action potential discharge: only small amounts of excitatory synaptic current can be inhibited. Moderate amounts of excitatory current always result in action potential discharge, despite AIS inhibition. Inhibition of the somadendrite by basket cells enhances the effect of AIS inhibition and vice versa. Thus the axoaxonic cells may act synergistically with basket cells: the AIS inhibition increases the threshold for action potential discharge, the basket cells then control the suprathreshold discharge.

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28. Feature Linking via Synchronization among Distributed Assemblies: Simulations of Results from Cat Visual Cortex

Reinhard Eckhorn, H. J. Reitboeck, M. Arndt, Peter W. Dicke

https://doi.org/10.1162/neco.1990.2.3.293 | pp. 293-307

We recently discovered stimulus-specific interactions between cell assemblies in cat primary visual cortex that could constitute a global linking principle for feature associations in sensory and motor systems: stimulus-induced oscillatory activities (35-80 Hz) in remote cell assemblies of the same and of different visual cortex areas mutually synchronize, if common stimulus features drive the assemblies simultaneously. Based on our neurophysiological findings we simulated feature linking via synchronizations in networks of model neurons. The networks consisted of two one-dimensional layers of neurons, coupled in a forward direction via feeding connections and in lateral and backward directions via modulatory linking connections. The models' performance is demonstrated in examples of region linking with spatiotemporally varying inputs, where the rhythmic activities in response to an input, that initially are uncorrelated, become phase locked. We propose that synchronization is a general principle for the coding of associations in and among sensory systems and that at least two distinct types of synchronization do exist: stimulus-forced (event-locked) synchronizations support “crude instantaneous” associations and stimulus-induced (oscillatory) synchronizations support more complex iterative association processes. In order to bring neural linking mechanisms into correspondence with perceptual feature linking, we introduce the concept of the linking

field (association field) of a local assembly of visual neurons. The linking field extends the concept of the invariant receptive field (RF) of single neurons to the flexible association of RFs in neural assemblies.

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29. Towards a Theory of Early Visual Processing

Joseph J. Atick, Amanda Redlich

https://doi.org/10.1162/neco.1990.2.3.308 | pp. 308-320

We propose a theory of the early processing in the mammalian visual pathway. The theory is formulated in the language of information theory and hypothesizes that the goal of this processing is to recode in order to reduce a “generalized redundancy” subject to a constraint that specifies the amount of average information preserved. In the limit of no noise, this theory becomes equivalent to Barlow's redundancy reduction hypothesis, but it leads to very different computational strategies when noise is present. A tractable approach for finding the optimal encoding is to solve the problem in successive stages where at each stage the optimization is performed within a restricted class of transfer functions. We explicitly find the solution for the class of encodings to which the parvocellular retinal processing belongs, namely linear and nondivergent transformations. The solution shows agreement with the experimentally observed transfer functions at all levels of signal to noise.

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30. Derivation of Linear Hebbian Equations from a Nonlinear Hebbian Model of Synaptic Plasticity

Kenneth D. Miller

https://doi.org/10.1162/neco.1990.2.3.321 | pp. 321-333

A linear Hebbian equation for synaptic plasticity is derived from a more complex, nonlinear model by considering the initial development of the difference between two equivalent excitatory projections. This provides a justification for the use of such a simple equation to model activity-dependent neural development and plasticity, and allows analysis of the biological origins of the terms in the equation. Connections to previously published models are discussed.

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31. Spontaneous Development of Modularity in Simple Cortical Models

Alex Chernjavsky, John Moody

https://doi.org/10.1162/neco.1990.2.3.334 | pp. 334-354

The existence of modular structures in the organization of nervous systems (e.g., cortical columns, patches of neostriatum, and olfactory glomeruli) is well known. However, the detailed dynamic mechanisms by which such structures develop remain a mystery. We propose a mechanism for the formation of modular structures that utilizes a combination of intrinsic network dynamics and Hebbian learning. Specifically, we show that under certain conditions, layered networks can support spontaneous localized activity patterns, which we call collective excitations, even in the absence of localized or spatially correlated afferent stimulation. These collective excitations can then induce the formation of modular structures in both the afferent and lateral connections via a Hebbian learning mechanism. The networks we consider are spatially homogeneous before learning, but the spontaneous emergence of localized collective excitations and the consequent

development of modules in the connection patterns break translational symmetry. The essential conditions required to support collective excitations include internal units with sufficiently high gains and certain patterns of lateral connectivity. Our proposed mechanism is likely to play a role in understanding more complex (and more biologically realistic) systems.

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32. The Bootstrap Widrow-Hoff Rule as a Cluster-Formation Algorithm

Geoffrey E. Hinton, Steven J. Nowlan

https://doi.org/10.1162/neco.1990.2.3.355 | pp. 355-362

An algorithm that is widely used for adaptive equalization in current modems is the “bootstrap” or “decision-directed” version of the Widrow-Hoff rule. We show that this algorithm can be viewed as an unsupervised clustering algorithm in which the data points are transformed so that they form two clusters that are as tight as possible. The standard algorithm performs gradient ascent in a crude model of the log likelihood of generating the transformed data points from two gaussian distributions with fixed centers. Better convergence is achieved by using the exact gradient of the log likelihood.

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33. The Effects of Precision Constraints in a Backpropagation Learning Network

P.W. Hollis, John S. Harper, J.J. Paulos

https://doi.org/10.1162/neco.1990.2.3.363 | pp. 363-373

This paper presents a study of precision constraints imposed by a hybrid chip architecture with analog neurons and digital backpropagation calculations. Conversions between the analog and digital domains and weight storage restrictions impose precision limits on both analog and digital calculations. It is shown through simulations that a learning system of this nature can be implemented in spite of limited resolution in the analog circuits and using fixed point arithmetic to implement the backpropagation algorithm.

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34. Exhaustive Learning

Daniel Schwartz, Vijay K. Samalam, Sara A. Solla, J. S. Denker

https://doi.org/10.1162/neco.1990.2.3.374 | pp. 374-385

Exhaustive exploration of an ensemble of networks is used to model learning and generalization in layered neural networks. A simple Boolean learning problem involving networks with binary weights is numerically solved to obtain the entropy S m and the average generalization ability G m as a function of the size m of the training set. Learning curves G m vs m are shown to depend solely on the distribution of generalization abilities over the ensemble of networks. Such distribution is determined prior to learning, and provides a novel theoretical tool for the prediction of network performance on a specific task.

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35. A Method for Designing Neural Networks Using Nonlinear Multivariate Analysis: Application to Speaker-Independent Vowel Recognition

Toshio Irino, Hideki Kawahara
https://doi.org/10.1162/neco.1990.2.3.386 | pp. 386-397

A nonlinear multiple logistic model and multiple regression analysis are described as a method for determining the weights for two-layer networks and are compared to error backpropagation. We also provide a method for constructing a three-layer network whose semilinear middle units are primarily provided to discriminate two categories. Experimental results on speaker-independent vowel recognition show that both multivariate methods provide stable weights with fewer iterations than backpropagation training started with random initial weights, but with slightly inferior performance. Backpropagation training with initial weights determined by a multiple logistic model after introduction of data distribution information gives a recognition rate of 98.2%, which is significantly better than average backpropagation with random initial weights.

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36. How to Solve the N Bit Encoder Problem with Just Two Hidden Units

Leonid Kruglyak
https://doi.org/10.1162/neco.1990.2.4.399 | pp. 399-401

I demonstrate that it is possible to construct a three-layer network of the standard feedforward architecture that can solve the N input-N output encoder problem with just two hidden units.

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37. Small Depth Polynomial Size Neural Networks

Zoran Obradovi■, Peiyuan Yan
https://doi.org/10.1162/neco.1990.2.4.402 | pp. 402-404

For polynomially bounded weights and polylogarithmic precision, analog neural networks of polynomial size and depth 3 are strictly more powerful than those of polynomial size and depth 2.

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38. Guaranteed Learning Algorithm for Network with Units Having Periodic Threshold Output Function

Mark Brady
https://doi.org/10.1162/neco.1990.2.4.405 | pp. 405-408

December 01 1990 Guaranteed Learning Algorithm for Network with Units Having Periodic Threshold Output Function In Special Collection: CogNet Mark J. Brady Mark J. Brady 3M Software and Electronics Resource, St. Paul, MN 55144-2000 USA Search for other works by this author on: This Site Google Scholar Author and Article Information Mark J. Brady 3M Software and Electronics Resource, St. Paul, MN 55144-2000 USA Online Issn: 1530-888X Print Issn: 0899-7667 © 1990 Massachusetts Institute of Technology1990 Neural Computation (1990) 2 (4): 405–408. https://doi.org/10.1162/neco.1990.2.4.405 Cite Icon Cite Permissions Share Icon Share Facebook Twitter LinkedIn MailTo Views Icon Views Article contents Figures & tables Video Audio Supplementary Data Peer Review Search Site Citation Mark J. Brady; Guaranteed Learning Algorithm for Network with Units Having Periodic Threshold Output Function. Neural Comput 1990; 2 (4): 405–408. doi: https://doi.org/10.1162/neco.1990.2.4.405 Download citation file:

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39. Active Perception and Reinforcement Learning

Steven D. Whitehead, Dana H. Ballard

https://doi.org/10.1162/neco.1990.2.4.409 | pp. 409-419

This paper considers adaptive control architectures that integrate active sensorimotor systems with decision systems based on reinforcement learning. One unavoidable consequence of active perception is that the agent's internal representation often confounds external world states. We call this phenomenon perceptual aliasing and show that it destabilizes existing reinforcement learning algorithms with respect to the optimal decision policy. A new decision system that overcomes these difficulties is described. The system incorporates a perceptual subcycle within the overall decision cycle and uses a modified learning algorithm to suppress the effects of perceptual aliasing. The result is a control architecture that learns not only how to solve a task but also where to focus its attention in order to collect necessary sensory information.

40. Structure from Motion with Impaired Local-Speed and Global Motion-Field Computations

Lucia M. Vaina, Norberto M. Grzywacz, Marjorie LeMay

https://doi.org/10.1162/neco.1990.2.4.420 | pp. 420-435

Humans can recover the three-dimensional structure of moving objects from their changing two-dimensional retinal image, in the absence of other cues to three-dimensional structure (Wallach and O'Connell 1953; Braunstein 1976). In this paper, we describe a patient, A.F., with bilateral lesions involving the visual cortex who is severely impaired on computing local-speed and global-motion fields, but who can recover structure from motion. The data suggest that although possibly useful, global-motion fields are not necessary for deriving structure from motion. We discuss these results from the perspective of theoretical models for this computation.

41. Learning Virtual Equilibrium Trajectories for Control of a Robot Arm

Reza Shadmehr

https://doi.org/10.1162/neco.1990.2.4.436 | pp. 436-446

The cerebellar model articulation controller (CMAC) (Albus 1975) is applied for learning the inverse dynamics of a simulated two joint, planar arm. The actuators were antagonistic muscles, which acted as feedback controllers for each joint. We use this example to demonstrate some limitations of the control paradigm used in earlier applications of the CMAC (e.g., Miller et al. 1987, 1990): the CMAC learns dynamics of the arm and not those of the feedback system. We suggest an alternate approach, one in which the CMAC learns to manipulate the feedback controller's input, producing a virtual trajectory, rather the

controller's output, which is torque. Several experiments are performed that suggest that the CMAC learns to compensate for the dynamics of the plant, as well as the controller.

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42. Discovering the Structure of a Reactive Environment by Exploration

Michael C. Mozer, Jonathan Bachrach

https://doi.org/10.1162/neco.1990.2.4.447 | pp. 447-457

Consider a robot wandering around an unfamiliar environment, performing actions and observing the consequences. The robot's task is to construct an internal model of its environment, a model that will allow it to predict the effects of its actions and to determine what sequences of actions to take to reach particular goal states. Rivest and Schapire (1987a,b; Schapire 1988) have studied this problem and have designed a symbolic algorithm to strategically explore and infer the structure of “finite state” environments. The heart of this algorithm is a clever representation of the environment called an update graph. We have developed a connectionist implementation of the update graph using a highly specialized network architecture. With backpropagation learning and a trivial exploration strategy — choosing random actions — the connectionist network can outperform the Rivest and Schapire algorithm on simple problems. Our approach has additional virtues, including the fact that the network can accommodate stochastic environments and that it suggests generalizations of the update graph representation that do not arise from a traditional, symbolic perspective.

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43. Computing with Arrays of Coupled Oscillators: An Application to Preattentive Texture Discrimination

Pierre Baldi, Ronny Meir

https://doi.org/10.1162/neco.1990.2.4.458 | pp. 458-471

Recent experimental findings (Gray et al. 1989; Eckhorn et al. 1988) seem to indicate that rapid oscillations and phase-lockings of different populations of cortical neurons play an important role in neural computations. In particular, global stimulus properties could be reflected in the correlated firing of spatially distant cells. Here we describe how simple coupled oscillator networks can be used to model the data and to investigate whether useful tasks can be performed by oscillator architectures. A specific demonstration is given for the problem of preattentive texture discrimination. Texture images are convolved with different sets of Gabor filters feeding into several corresponding arrays of coupled oscillators. After a brief transient, the dynamic evolution in the arrays leads to a separation of the textures by a phase labeling mechanism. The importance of noise and of long range connections is briefly discussed.

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44. Robust Classifiers without Robust Features

Alan J. Katz, Michael T. Gately, D.R. Collins

https://doi.org/10.1162/neco.1990.2.4.472 | pp. 472-479

We develop a two-stage, modular neural network classifier and apply it to an automatic target recognition problem. The data are features extracted from infrared and TV images. We discuss the problem of robust classification in terms of a family of decision surfaces, the members of which are functions of a set of global variables. The global variables characterize how the feature space changes from one image to the next. We

obtain rapid training times and robust classification with this modular neural network approach.

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45. Use of an Artificial Neural Network for Data Analysis in Clinical Decision-Making: The Diagnosis of Acute Coronary Occlusion

William G. Baxt

https://doi.org/10.1162/neco.1990.2.4.480 | pp. 480-489

A nonlinear artificial neural network trained by backpropagation was applied to the diagnosis of acute myocardial infarction (coronary occlusion) in patients presenting to the emergency department with acute anterior chest pain. Three-hundred and fifty-six patients were retrospectively studied, of which 236 did not have acute myocardial infarction and 120 did have infarction. The network was trained on a randomly chosen set of half of the patients who had not sustained acute myocardial infarction and half of the patients who had sustained infarction. It was then tested on a set consisting of the remaining patients to which it had not been exposed. The network correctly identified 92% of the patients with acute myocardial infarction and 96% of the patients without infarction. When all patients with the electrocardiographic evidence of infarction were removed from the cohort, the network correctly identified 80% of the patients with infarction. This is substantially better than the performance reported for either physicians or any other analytical approach.

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46. An Efficient Gradient-Based Algorithm for On-Line Training of Recurrent Network Trajectories

Ronald J. Williams, Jing Peng

https://doi.org/10.1162/neco.1990.2.4.490 | pp. 490-501

A novel variant of the familiar backpropagation-through-time approach to training recurrent networks is described. This algorithm is intended to be used on arbitrary recurrent networks that run continually without ever being reset to an initial state, and it is specifically designed for computationally efficient computer implementation. This algorithm can be viewed as a cross between epochwise backpropagation through time, which is not appropriate for continually running networks, and the widely used on-line gradient approximation technique of truncated backpropagation through time.

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47. Convergence Properties of Learning in ART1

Michael Georgiopoulos, Gregory L. Heileman, Juxin Huang

https://doi.org/10.1162/neco.1990.2.4.502 | pp. 502-509

We consider the ART1 neural network architecture. It is shown that in the fast learning case, an ART1 network that is repeatedly presented with an arbitrary list of binary input patterns, self-stabilizes the recognition code of every size-l pattern in at most l list presentations.

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48. A Polynomial Time Algorithm That Learns Two Hidden Unit Nets

Eric B. Baum

https://doi.org/10.1162/neco.1990.2.4.510 | pp. 510-522

Let N be the class of functions realizable by feedforward linear threshold nets with n input units, two hidden units each of zero threshold, and an output unit. This class is also essentially equivalent to the class of intersections of two open half spaces that are bounded by planes through the origin. We give an algorithm that probably almost correctly (PAC) learns this class from examples and membership queries. The algorithm runs in time polynomial in n, ■ (the accuracy parameter), and δ (the confidence parameter). If only examples are allowed, but not membership queries, we give an algorithm that learns N in polynomial time provided that the probability distribution D from which examples are chosen satisfies D(x) = D(–x) ∀x. The algorithm yields a hypothesis net with two hidden units, one linear threshold and the other quadratic threshold.

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49. Phase Transitions in Connectionist Models Having Rapidly Varying Connection Strengths

James A. Reggia, Mark Edwards

https://doi.org/10.1162/neco.1990.2.4.523 | pp. 523-535

A phase transition in a connectionist model refers to a qualitative change in the model's behavior as parameters determining the spread of activation (gain, decay rate, etc.) pass through certain critical values. As connectionist methods have been increasingly adopted to model various problems in neuroscience, artificial intelligence, and cognitive science, there has been an increased need to understand and predict these phase transitions to assure meaningful model behavior. This paper extends previous results on phase transitions to encompass a class of connectionist models having rapidly varying connection strengths (“fast weights”). Phase transitions are predicted theoretically and then verified through a series of computer simulations. These results broaden the range of connectionist models for which phase transitions are identified and lay the foundation for future studies comparing models with rapidly varying and slowly varying connection strengths.

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50. An M-ary Neural Network Model

Richard Bijjani, Pankaj Kumar Das

https://doi.org/10.1162/neco.1990.2.4.536 | pp. 536-551

An M-ary neural network model is described, and is shown to have a higher error correction capability than the bipolar Hopfield neural net. The M-ary model is then applied to an image recognition problem.

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51. Index to Volume 2

(authors unknown)

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No abstract available.